CSC1058 USER INTERFACE DESIGN & IMPLEMENTATION

MODULE COORDINATOR & LECTURER
Hyowon Lee

TEACHING ASSISTANT
Haoyan Zhang

	Tuesday (3 – 4pm)	Wednesday (11am - 12pm)
Week 1 (7 Sep)	Introduction/ orientation	HCI background
Week 2 (14 Sep)	Design guidelines	Design guidelines
Week 3 (21 Sep)	Design guidelines	Design guidelines
Week 4 (28 Sep)	Design guidelines	Project brainstorming
Week 5 (5 Oct)	Guidelines recap	Guidelines recap
Week 6 (12 Oct)	Preventing negative impact	Usability evaluation
Week 7 (19 Oct)	Usability evaluation	Usability evaluation
Week 8 (26 Oct)	Usability evaluation	Usability evaluation
Week 9 (2 Nov)	Usability evaluation	Usability evaluation
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Week 11 (16 Nov)	Project discussion	Project feedback
Week 12 (23 Nov)	Summary/ reflection	Summary/ reflection

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Heuristic Inspection

- Purpose: find out usability bugs
- Design expert goes through the UI and finds the bugs against guidelines

  e.g. checking a newly-designed website… writing style is inconsistent! (“Be consistent”)

  - Reasonably simple and short
  - Depends on the inspector’s capability…

    More than 1 inspector?

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Collaborative

ACTIVITY (15 min)
HEURISTIC INSPECTION

First, in pair:

1. Choose a commonly-used app or website (write down)
2. Decide a typical task of that app/website (write down)

Second, individually:

3. Go through the task, step by step, while checking on our design guideline card)
4. At each step, write down where the design violates the guidelines

Then in pair:

5. Compare with your partner, fill in the table

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DISCUSSION
Example UIs

Lifelogging archive UI
Journalist video clip search UI

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Interview & Questionnaire

- Obtains "why"
- Needs interview skill, questionnaire design skill
- **Interpret** the response
- Very usefully:
  - Interaction logging analysis + follow-up interview/questionnaire

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Designing a questionnaire

- Make questions clear and specific
- When possible, ask closed questions and offer a range of answers
- General questions first
- Avoid complex questions
- When scale is used, make sure they don't overlap
- Scale should be consistent (e.g. 1 as small)
- Trade-off: length of the questionnaire

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What is your age? _____

Most participants feel reluctant to enter the exact age – use range…

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What is your age?

- ❑ < 20
- ❑ 20 - 25
- ❑ 25 - 30
- ❑ 30 - 35
- ❑ 35 - 40
- ❑ 40 - 45
- ❑ > 45

Make sure no overlap of values

What is your age?

- ❑ ≤ 20
- ❑ 21 - 25
- ❑ 26 - 30
- ❑ 31 - 35
- ❑ 36 - 40
- ❑ 41 - 45
- ❑ > 45

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(Likert scale)

The use of colour is excellent (where 1 represents strongly disagree and 5 represents strongly agree)

❑1 ❑ 2 ❑ 3 ❑ 4 ❑ 5

9

(Semantic differential scale)

For each pair of adjectives, place a cross at the point between them that reflects the extent to which you believe the adjectives describe the UI. You should place only one cross on each line.

- Attractive — Ugly
- Clear — Confusing
- Dull — Colourful
- Exciting — Boring
- Annoying — Pleasing
- Helpful — Unhelpful
- Poorly designed — Well designed

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Ready-made questionnaires

- **QUIS**: Quesionnaire for User Interaction Satisfaction (U of Maryland)
  https://www.uxtesttools.com/tools/quis (from UX Test Tools)
  SUMI: Software Usability Measurement Inventory
  https://www.uxtesttools.com/tools/sumi (from UX Test Tools)
  SUS: "Quick and Dirty" System Usability Scale
  https://measuringu.com/sus/ (from MeasuringU)

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User feedback after deployment

- Simple and voluntary
  - Dissatisfied users will respond
  - Dedicated users will respond
- Don't just collect them, but:
  - Categorise
  - Determine patterns
  - Find root cause of issues

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Lab testing

"Usability Laboratory"





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Lab testing

"Usability Laboratory"



- Expensive to build
  - Room(s)
  - Equipment (cameras, monitors, software (e.g. Morae)…
- Expensive to run
  - Rental of room facility
  - Careful plan
  - Paying the test users

→ When measuring or comparing quantity is important

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"Usability Laboratory"

… but what about mobile or home applications?

Usability Lab seems to imply a **limited vision**, because…

Interacting with computer ≠ using a desktop PC

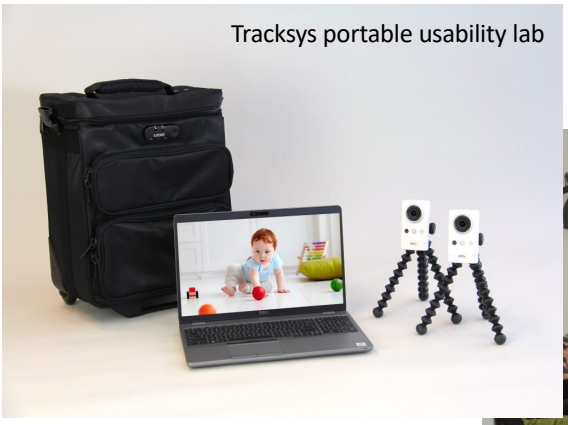
Lab testing

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- Expensive to run
  - Rental of room facility
  - Careful plan
  - Paying the test users

→ When measuring or comparing quantities is important

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15





Bunnyfoot large observation room

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EVALUATION TECHNIQUES: SUMMARY

Asking Experts
- Heuristic Inspection

Observing Users
- Indirect Observation (Interaction log[illegible]ng, diary, etc.)
- Direct Observation

Asking Users
- Think-aloud
- Interview/Questionniare
- User Feedback

Asking & Observing Users
- Lab Testing

LESS → MORE
Intrusiveness to users

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Asking Experts
- Heuristic Inspection

Observing Users
- Indirect Observation (
- Direct Observation

Asking Users
- Think-aloud
- Interview/Question
- User Feedback

Asking & Observing Users
- Lab Testing



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Ethics of User Testing

"Users" are **human beings**

Human subjects have been seriously abused in the past…

- Nazi concentration camps (1940-1945)
- Tuskegee Institute syphilis study (1932-1972)
- MIT Fernald School study (1940s – 1950s)
- Milgram experiment at Yale (1960s)

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Ethics of User Testing

"Users" are hu

Human s
seriously

- Nazi conc
- Tuskegee
- MIT Fern
- Milgram

Belmont Report (1979)
http://www.hhs.gov/ohrp/humansubjects/guidance/belmont.html

Respect for persons
- Voluntary participation
- Informed consent
- Protection of vulnerable population (children, people with disabilities, etc.)

Beneficence
- Do no harm
- Risk vs. benefit

Justice

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Ethics of User Testing

Institutional Review Board

- **Ethics Review**
- All research involving human subjects requires approval from DCU ethics review

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Ethics of User Testing

Institutional Review Board

Pressure on the user…

- Performance anxiety
- Comparing self to other subjects
- Feeling stupid in front of the observer
- Competing with other subjects

view

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Ethics of User Testing

Treat users with respect

- Don't waste their time
- Make the user comfortable
- Inform the user as fully as possible
- Preserve the user's privacy
- The user can stop at any time

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Ethics of User Testing

Treat users with resp

- Don't waste their tim
- Make the user comfo
- Inform the user as ful possible
- Preserve the user's pri
- The user can stop at ar

"Keep in mind that we are testing the UI, we're not testing you"

"The system is likely to have some problems that makes it difficult to use. We needs your help to find these problems"

"Your test result will be completely confidential"

"You have the right to stop and leave at any point"

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REPORTS…

- Investigate if there are similar approaches, systems, UIs, products that do "it"
- Find out how the issue has been framed or tackled in other studies/products
- Find and reference related work - specific work related to the specific focus of the project, not general UX-related work

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REPORTS…

"intuitive"
"easy to use"
"user-friendly"
"clean"
:

"Is the **mapping** properly done?"
"It's an issue of making the functionality more **visible**"
"Of the **usability criteria**, which is the priority? Are we going for **efficiency** or **learnability**?"
"How can we reduce the **click distance** here?"

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REPORTS…

"intuitive"
"easy to use"
"user-friendly"
"clean"
:

"Who are the **target users**?"
"Try an informal **observation** session with a few test users to find some obvious **usability bugs**"
"Does the app provide any **accessibility** features?"
"Not a great **metaphor**…"
"A simple matching algorithm could offer some powerful **short-cuts** here"

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REPORTS…

"intuitive"
"easy to use"
"user-friendly"
"clean"
:

"Contents on this page is not **sharpened** enough"
"Well, a very poor **affordance** here"
"Can we **prevent the error** in the first place?
"Widgets are not visually **aligned** well"

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Evaluation Metrics

- **Measures** - what
- **Techniques** - how

Methodology on how to capture the Measures

For example, evaluate a new mobile game app

Measure: level of user satisfaction

Technique: Questinnaire

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Example: Website evaluation

"Evaluation Criteria"

e.g. Annual Webby Awards

http://www.webbyawards.com

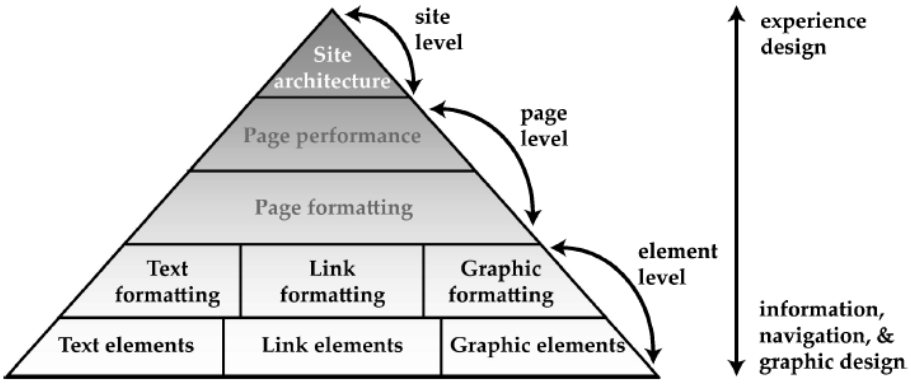
- Content
- Structure and Navigation
- Visual Design
- Functionality
- Interactivity
- Innovation
- Overall Experience

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Example: Website evaluation

Conceptual model of web interface & framework for evaluation



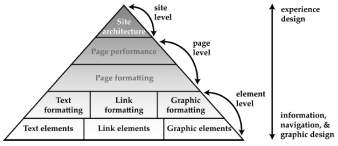
Ivory, M. and Megraw, R. (University of Washington) Evolution of Web Site Design Patterns. ACM Transactions on Information Systems, 23(4), 2005. pp463-497.

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Example: Website evaluation

Conceptual model of web interface & framework for evaluation



Category	Number of Measures	Aspects Measured
Text elements	31	Amount of text, type, quality, and complexity. Includes visible and invisible text.
Link elements	6	Number and type of links.
Graphic elements	6	Number and type of images.
Text formatting	24	How body text is emphasized; whether some underlined text is not in text links; how text areas are highlighted; font styles and sizes; number of text colors; number of times text is repositioned.
	3	Colors used for links and whether there are text links that are not underlined.
matting	7	Minimum, maximum, and average image width and height; page area covered by images.
Page formatting	27	Color use, fonts, page size, use of interactive elements, page style control, and so on. Key measures include evaluating the quality of color combinations (for text and panels) and predicting the functional type of a page.[a]
Page performance	37	Page download speed; page accessibility for people who have impairments; presence of HTML errors; and "scent" strength.[b]
Site architecture	16	Consistency of page elements, element formatting, page formatting and performance, and site size (number of pages or documents).[c]

Total 157 measures!

Ivory, M. and Megraw, R. (University of Washington) Evolution of Web Site Design Patterns. ACM Transactions on Information Systems, 23(4), 2005. pp463-497.

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Usability Requirements

Captures goals and associated measures for a UI

Make a list:

(i) Usability goals
(ii) User experience goals

Then prioritise them

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Usability Requirements
(i) Usability goals

- Speed of performance = Efficiency
- Learning how to use = Learnability
- Retention over time = Memorability
- Rate of errors by user = Error rate
- Subjective satisfation = Satisfaction

"Usability goals"
"Usability criteria"
"Usability factors"
:

Prioritise!

5 Usability Criteria
- Efficiency
- Learnability
- Memorability
- Error rate
- Satisfaction

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Usability Requirements
(ii) UX goals

- Satisfying
- Enjoyable
- Motivating
- Fun
- Aesthetically pleasing
- Emotionally fulfilling
- Reflectively appropriat
- ⋮

Prioritise!

5 Usability Criteria
- Efficiency
- Learnability
- Memorability
- Error rate
- Satisfaction

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So…

(i) Usability goals

- Efficiency
- **Learnability**
- Memorability
- Error rate
- Satisfaction

(ii) UX goals

- Satisfying
- Enjoyable
- **Motivating**
- Fun
- Aesthetically pleasing

⋮

Prioritise!

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(i) Usability goals

- Efficiency
- Learnability
- Memorability
- Error rate
- Satisf

Prioritise!

(ii) UX go

- Satisfy
- Enjoyable
- Motivating
- Fun
- Aesthetic

⋮

Example 1:

Word UI for writing a letter

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Example 1:

Word UI for writing a letter

(i) Usability goals

- **Efficiency** 1
- Learnability
- Memorability
- **Error rate** 2
- Satisfaction

(ii) UX goals

- Satisfying
- **Enjoyable** 3
- Motivating
- Fun
- Aesthetic
- ⋮

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Example 1:

Word UI for writing a letter

(i) Usability goals

- **Efficiency** 1
- Learnability
- Memorability
- **Error rate** 2
- Satisfaction

(ii) UX goals

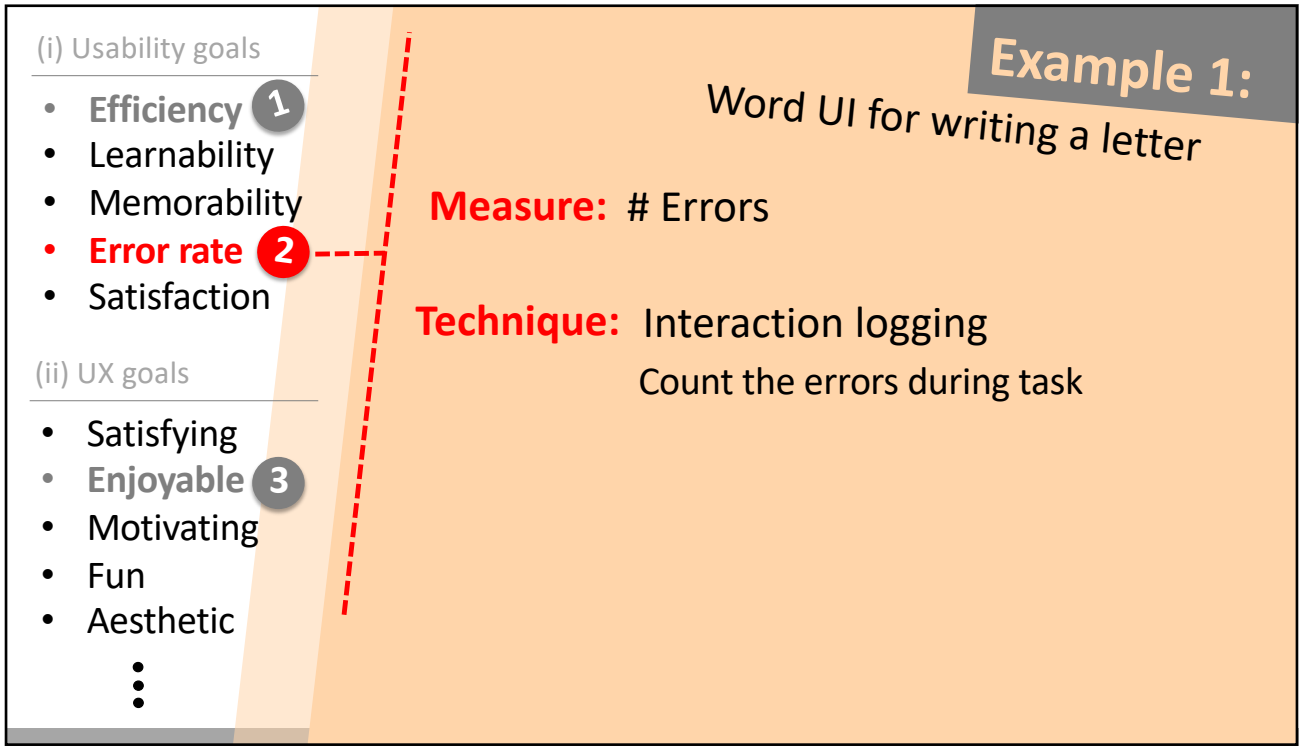
- Satisfying
- **Enjoyable** 3
- Motivating
- Fun
- Aesthetic
- ⋮

Measure: Time taken (minutes)

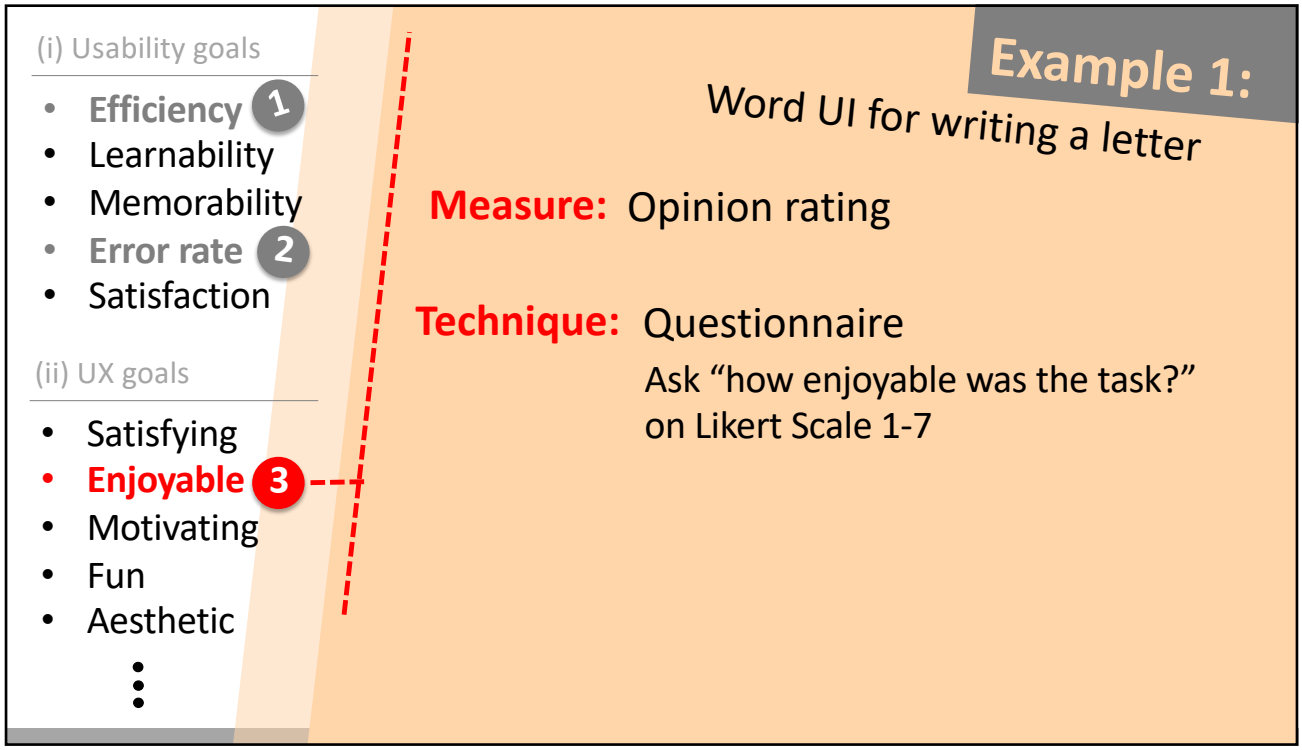
Technique: Lab testing

Task: write a 1-page letter to your aunt to inform about the wedding last week

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(i) Usability goals
Efficiency 1
Learnability
Memorability
Error rate 2
Satisfaction
(ii) UX goals
Satisfying
Enjoyable 3
Motivating
Fun
Aesthetic
Example 1:
Word UI for writing a letter
Measure: # Errors
Technique: Interaction logging
Count the errors during task

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(i) Usability goals
Efficiency 1
Learnability
Memorability
Error rate 2
Satisfaction
(ii) UX goals
Satisfying
Enjoyable 3
Motivating
Fun
Aesthetic
Example 1:
Word UI for writing a letter
Measure: Opinion rating
Technique: Questionnaire
Ask "how enjoyable was the task?"
on Likert Scale 1-7

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